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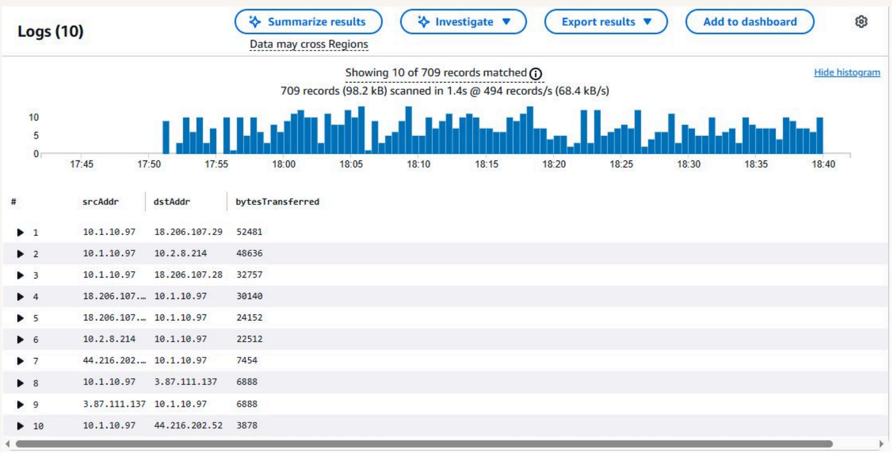
VPC Monitoring with Flow Logs

M Mohammed Dawoud Mota



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a secure, isolated cloud network that lets you control IP ranges, subnets, routes, and security settings, making it easier to protect and manage AWS resources.

How I used Amazon VPC in this project

I used Amazon VPC to create a secure network environment, set up subnets, and control traffic using route tables and security groups to ensure safe communication between resources.

One thing I didn't expect in this project was...

I didn’t expect how detailed network configurations could be, especially managing subnets, route tables, and security groups to ensure proper communication between resources.

This project took me...

This project took me 40 minutes to complete.

In the first part of my project...

Step 1 - Set up VPCs

In this step, I will create two VPCs from scratch. So I can revise my learning from previous parts of the project.

Step 2 - Launch EC2 instances

In this step, I will launch EC2 instances in each VPC so that they can send data to each other and communicate.

Step 3 - Set up Logs

In this step, I will use VPC flowlogs to set up a way to log all inbound and outbound network traffic. I will then set up a space that will store all those logs.

Step 4 - Set IAM permissions for Logs

In this step, I will give VPC Flow Logs the permission to write logs and send them to CloudWatch. And then I will finish setting up my subnet's flow log.

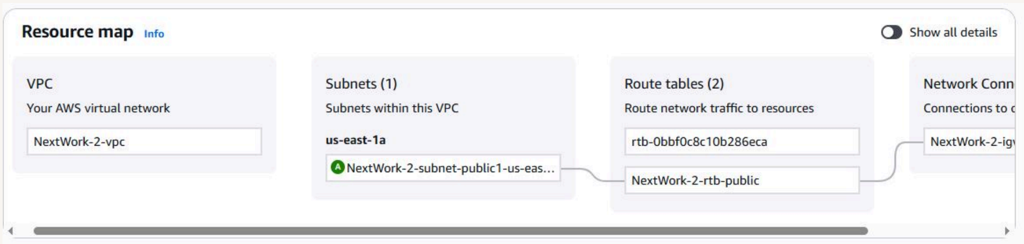
Multi-VPC Architecture

I started my project by launching a VPC which has one public subnet.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique so that resources within the VPC won't have overlapping IPs with each other. This will also make it to differentiate which resources belong to a VPC.

I also launched EC2 instances in each subnet

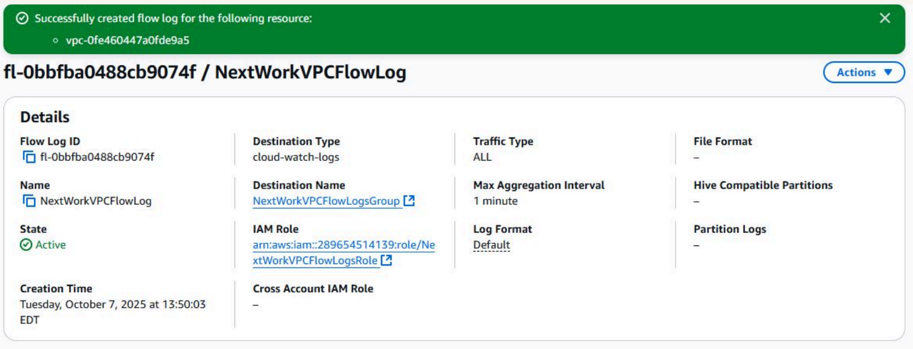
My EC2 instances' security groups allow SSH access from anywhere and ICMP access from anywhere.



Logs

Logs are like a diary for computer systems. Everything that happens is recorded. It is the place to go to when you want to know what is happening with your system, and keep an eye on what is happening.

Log groups are like a big folder in AWS that keeps all related logs together. Logs that share the same source usually go together, but they can also be categorized based on the region.

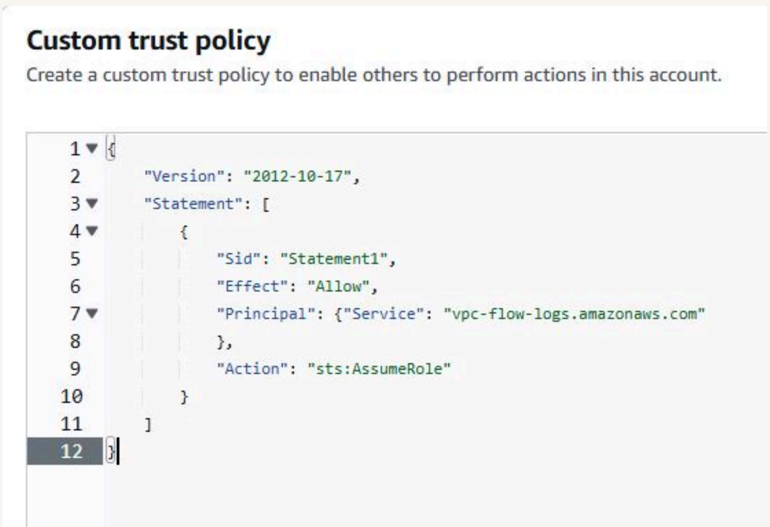


IAM Policy and Roles

I created an IAM policy because VPC Flow Logs by default don't have the permission to record logs and store them in my CloudWatch log group. This policy makes sure that my VPC can now send log data to my log group

I also created an IAM role because that it what is required for the VPC Flow Logs.

A custom trust policy is a specific type of policy! They're different from IAM policies. While IAM policies help you define the actions a user/service can or cannot do, custom trust policies are used to very narrowly define who can use a role.



In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step, I will get Instance 1 to send test messages to Instance 2 so that we can see if the flow logs are working.

Step 6 - Set up a peering connection

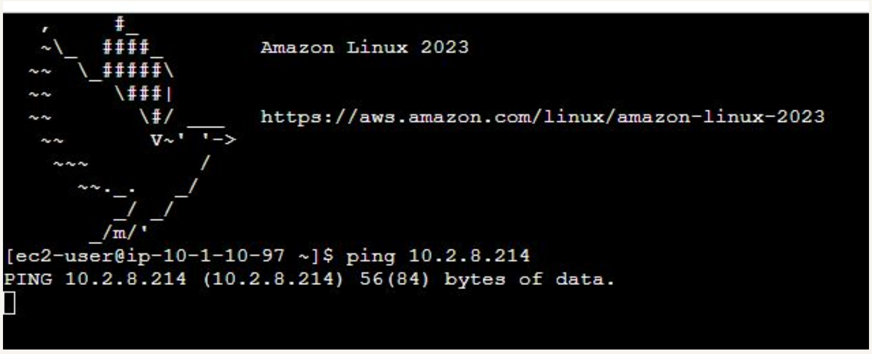
In this step, I will add a peering connection to bridge my two VPCs together.

Step 7 - Analyze flow logs

In this step, I will review the flow logs recorded about VPC 1's public subnet. And then I will analyze the flow logs to get some insights.

Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means there might be a problem with the connection. VPC 2's network might be blocking the traffic used with ping.



I could receive ping replies if I ran the ping test using the other instance's public IP address, which means Instance 2 is correctly configured to respond to ping requests. And Instance 1 can communicate with it across the public internet.

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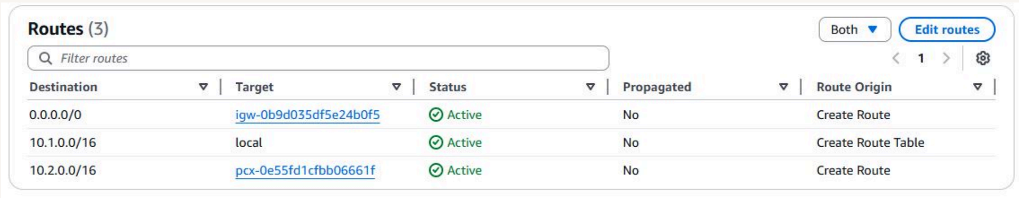
NextWork Student

Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because there was no peering connection between VPC 1 and VPC 2.

To solve this, I set up a peering connection between my VPCs

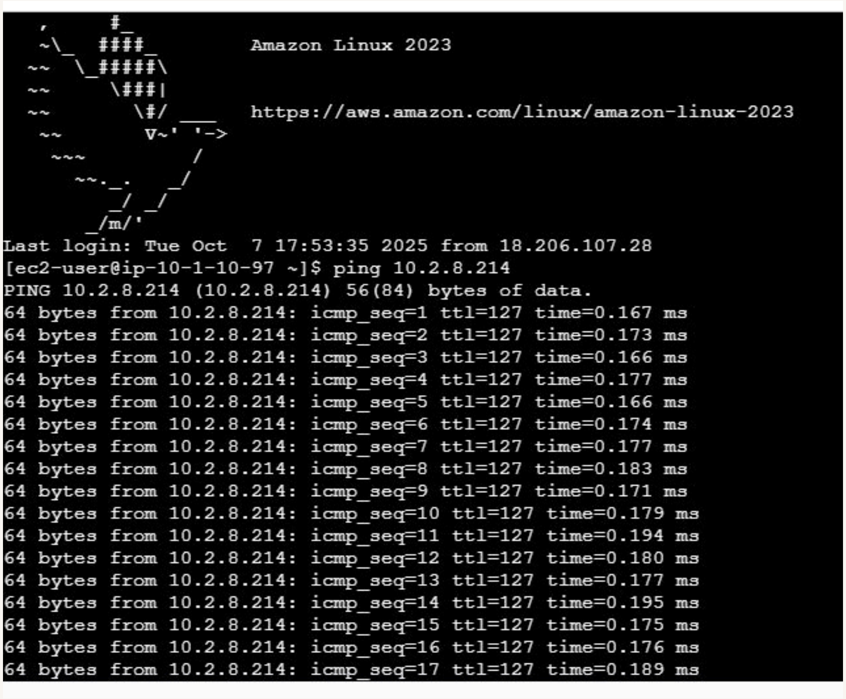
I also updated both VPCs' route tables so that a connection can be established between the two VPCs, and the traffic goes to the correct locations.



Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0b9d035df5e24b0f5	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	pcx-0e55fd1cfbb06661f	Active	No	Create Route

Connectivity troubleshooting

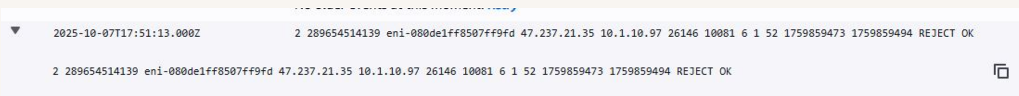
I received ping replies from Instance 2's private IP address! This means that the VPCs can communicate with each other and are able to send and receive data.



Analyzing flow logs

A flow log records details about network traffic, including source and destination IPs, ports, protocol, packets, bytes, timestamps, action (accept/reject), and log status, helping monitor and troubleshoot VPC network activity.

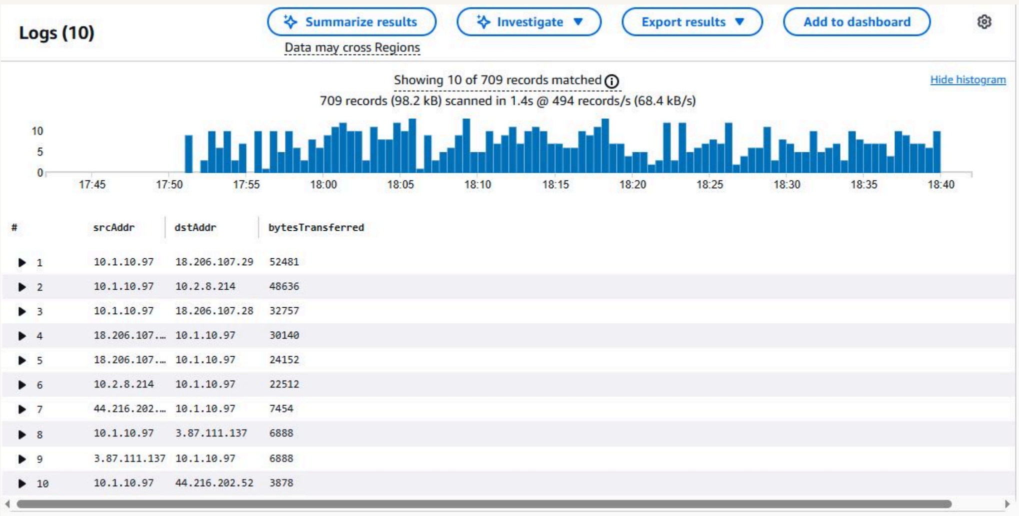
For example, the flow log I've captured tells us that the traffic was rejected.



Logs Insights

Logs Insights is a CloudWatch feature that analyzes your logs. In Log Insights, you use queries to filter, process, and combine data to help you troubleshoot problems or better understand your network traffic!

I ran the query "Top 10 byte transfers by source and destination IP addresses". This query analyzes the top 10 biggest data transfers between IP addresses in my network.



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